

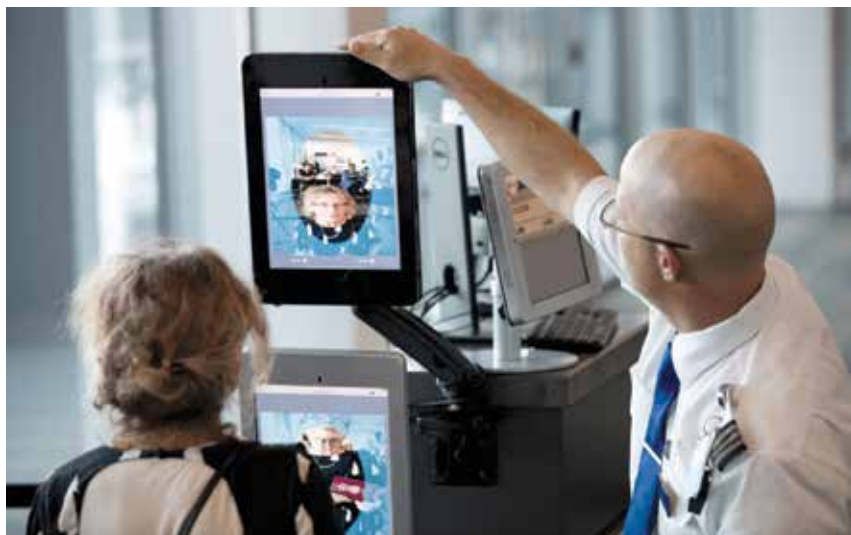
These technologies are designed to integrate seamlessly with existing air traffic control frameworks and communicate efficiently with nearby aircraft and ground stations. In parallel, Airbus is advancing autonomous aviation through initiatives such as Vahana and CityAirbus. Vahana is a single-passenger electric vertical take-off and landing (eVTOL) aircraft that has successfully completed multiple test flights. The programme focuses on integrating AI for vertical flight control, stability management, and route planning in densely populated urban environments. Similarly, CityAirbus is a multi-passenger air taxi designed to operate autonomously within defined urban air corridors.

Passenger experience and personalisation

Artificial intelligence is transforming the passenger experience in aviation by enabling a more seamless, personalised, and efficient journey from ticket purchase to final arrival. Through the integration of AI into customer service, operational processes, and biometric security systems, airlines and airports are significantly improving passenger convenience and satisfaction.

One prominent application is the automation of customer service. Major airlines such as Delta Air Lines have deployed AI-powered chatbots capable of resolving customer enquiries with a high degree of accuracy. These systems provide real-time support for flight changes, rebooking, seat selection, baggage tracking, and travel documentation. Delta reports that nearly 95% of customer queries are resolved without human intervention, resulting in faster response times and allowing support staff to focus on more complex cases.

JetBlue Airways has extended this approach by integrating generative AI into its customer service operations. By automating the generation of customised responses and service recommendations, JetBlue reportedly saved 73,000 hours of agent time in a single quarter,



Facial recognition at airports (Courtesy: aviationtoday.com)

demonstrating how generative AI can improve service efficiency while reducing operational costs.

Within airports, AI is reshaping identity verification and boarding processes. Facial recognition systems and biometric check-in solutions are increasingly adopted at major international hubs such as Singapore Changi, Heathrow, and Hartsfield-Jackson Atlanta International Airport. These technologies use AI-driven computer vision to rapidly match passengers' facial data with passport and ticket records, reducing dependence on physical documents and significantly cutting waiting times at security screening and boarding gates.

In addition, recommendation engines that analyse passenger travel history and preferences can suggest relevant add-ons, such as lounge access, priority boarding, or meal options, during booking and check-in stages. This level of personalisation enhances passenger engagement while improving overall service efficiency.

Weather prediction and impact analysis

Weather remains one of the most unpredictable and influential factors in aviation operations. Phenomena such as

turbulence, wind shear, thunderstorms, and fog can cause delays, route deviations, or flight cancellations. Artificial intelligence is reshaping aviation meteorology by enabling airlines and airports to anticipate and respond to changing weather conditions with far greater accuracy.

AI systems can analyse vast meteorological datasets—including satellite imagery, radar data, wind profiles, humidity levels, and atmospheric pressure—much faster than traditional forecasting tools. By applying machine learning techniques to identify complex patterns, AI improves both short-term and long-range weather forecasts, providing airlines with additional time to prepare operational responses.

A notable example is Lufthansa's collaboration at Zurich Airport, where AI-driven systems were deployed to improve wind pattern forecasting accuracy. By feeding real-time data into predictive models, Lufthansa achieved a reported 40% improvement in wind forecast precision, a critical factor during take-off and landing operations. This improvement supported better runway utilisation, reduced fuel burn during holding patterns, and minimised last-minute delays caused by unexpected wind shifts.